

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_f9255361-69d\agent-  
a082a2583ae9975f1.jsonl`  
- SHA-256: `ec145b5d06c6cd98ead09a7e474c334f40d155d4b95cc91946f26af87dd43ad0`  
- Source modified: `2026-06-22T09:35:02+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_f9255361-69d`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-22T09:32:46.024Z)

You are auditing ONE doc in the KhelSutra brand/site repo for staleness, as part of a "are all the docs up to date?" sweep.

Repo root: C:\Users\avidu\Projects\kheelsutra (bash path: /c/Users/avidu/Projects/kheelsutra)
The repo is on `master`, freshly synced to remote HEAD ? the latest merged PR is #94. Today is 2026-06-22.

DOC TO AUDIT: docs/DRY_RUN_CUJ.md

Your job: read the doc IN FULL, extract every concrete, falsifiable claim, and VERIFY each against the LIVE repo ? do NOT trust the doc's own wording. Hard discipline rule: a claim is "correct" ONLY when you have file:line or command-output evidence for it; if you cannot verify it, mark verdict "unverifiable" ? never guess, never assume the doc is right. Re-check against actual code/git.

Kinds of claims to check:

- File / path / route references ("served at /capture", "see src/store.js", "public/og.png", "systemd kheelsutra-site") -> does the path or route ACTUALLY exist now? Confirm with git ls-files / Read / the routing in site/server.js + site/src/index.js.
- PR numbers and counts ("PRs #1-#8 merged", "20 tests green", "6 languages", "3-step wizard", "P0..P5 done") -> verify against `git -C <root> log`, the actual test output/count, and the code.
- Status markers (DONE / pending / IN PROGRESS / DRAFT / "?" / "owner-gated" / "NEXT") -> still accurate vs the code & git history? Is something marked pending that is actually shipped, or vice-versa?
- Dates, "live at <url>", "merged in #N".
- Whether the doc describes something now SUPERSEDED or fully shipped -> is it an archive candidate (a tracker whose work is complete, or content duplicated by docs/archives/)? Note: docs/archives/FRIENDLY_AND_MULTILINGUAL.md already exists as the archived multilingual tracker.

Tools: Read the doc + every cited file; Grep/Glob to locate code; Bash for `git -C /c/Users/avidu/Projects/kheelsutra log --oneline`, `... ls-files`, and greps. To check a test count, you may run the suite (`cd /c/Users/avidu/Projects/kheelsutra/site && node --test` or `npm test`), but prefer cheap checks. Verify against master as it stands NOW.

Severity: high = a factually wrong claim a reader would act on (wrong path/route, a "DONE" that isn't, a wrong count/PR that misleads). medium = outdated-but-harmless (stale PR range, old "where we are", superseded framing). low = cosmetic.

Do NOT modify any file ? audit only. Return ONLY the structured object (doc, overallStatus, findings[], summary).

2. assistant (2026-06-22T09:32:48.467Z)

I'll audit the doc. Let me start by reading it in full and syncing the repo state.

3. user (2026-06-22T09:32:49.420Z)

```
1      # Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local
(multilingual)
2
3      > **Status:** `DRAFT` ? owner-gated . **Owner:** Avi Dullu . **Created:** 2026-06-21 .
**Goal for this session.**
4      > **Umbrella:** a milestone of `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the appliance) +
`docs/PER_RALLY_SELECTION.md` (the shipped SPA back-half).
5      > **Honesty note:** grounded in a file:line integration-surface map (this session);
claims tagged `[verified]` / `[gap]`.
6
7      ## 0. The goal (owner's words)
8      Operate the **web UI** ? point at a **video on the GCS bucket** ? the engine (tool)
processes it ? **review/select rallies in the UI** ? **compile** ? **download the reel to the
local machine**. Then **repeat with the UI in Hindi** to prove the multilingual workflow.
9
10     ## 1. What already works vs the gap `[verified]`
11     **? Back half (shipped ? PER_RALLY_SELECTION M0):** the SPA loads `?video=<id>` ? `POST
/api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ?
`<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works
today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns
`download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).
12
13     **? Front half ? the gap is the SPA, NOT the engine `[corrected 2026-06-21; see §8]`:**
14     - The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local),
can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in
the URL. **This SPA front-half is the *only* net-new work for Path B (= B-P2).**
15     - **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified
file:line]`:** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`)
`video_path` directly. Submit resolves the URI and **skips the local-existence check for non-
local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`/`gcs`
normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch
(`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same**
hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into
the API's SQLite `videos`/`play_segments`. Proven end-to-end by
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202,
not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the
hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
16     - *Where the earlier "net-new" claim went wrong:* it conflated the **API path**
(`/api/process`, which DID gain URI ingest in M2) with the separate **worker CLI** (`python -m
backend.worker infer --video-uri gcs://`, which writes `intervals.csv`/`report.json` to the
bucket ? **not** the API DB. The CLI distinction is real; the "the endpoint is local-only"
conclusion was not. ? **Engine-surface claims drift between sessions** (the engine ships PRs
independently) ? re-verify any `[gap]`/`net-new` engine claim against the cited `file:line`
before scoping it as work.
17
18     ## 2. Two paths to the goal
19     **Path A ? Cheapest dry run NOW (~zero new code).** Pre-process the bucket video out-of-
band to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the
**multilingual UI-driven workflow** end-to-end. The only non-UI step is "point at the bucket" (a
one-line CLI pre-process).
20
21     **Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first
scoped).** The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA
front-half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process`
? an **ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load
`result.video_id` (which the existing back-half already renders). **No net-new engine endpoint**
? B-P1 is already done; the remaining work is **B-P2 (the SPA)**).
22
23     ## 3. Path A runbook (the cheapest dry run) `[verified-steps]`
24     1. **Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`,
binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only, no
```

WSL** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)

25 2. **Get the bucket video local + into the API DB:**

26 `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST /api/process {"video\_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET /api/jobs/{job\_id}` until `result.video\_id`. *(`gemini` needs `GEMINI\_API\_KEY`; `motion` is key-free but hides fabricated metadata.)*

27 3. **Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`, same-origin) ? `http://localhost:5173/?video=<video\_id>` ? rallies auto-load.

28 4. **Select ? Build ? download:** pick rallies ? Build ? `<video>` preview ? click the download link ? MP4 saved local.

29 5. **Repeat in Hindi:** click `?????` (or `?lang=hi`) ? redo select?compile?download ? confirm the whole flow works with the UI in Hindi.

30

31 **Small unblocked SPA polish that makes Path A more "via the UI" (independent of A/B, useful in both): a **Load a video by ID** input (paste the `video\_id` ? load, instead of editing the URL) + consume the engine's `download\_url` (instead of hand-building it). One tight PR.

32

33 ## 4. Open decisions ? OWNER (these gate the dry run)

34 1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-only**); the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7), then B (B-P2, the SPA ingest/progress screen).**

35 2. **Compute / detector:** **Gemini** (real rallies; needs your `GEMINI\_API\_KEY`; uploads downsized chunks to Google = paid + cloud) · **motion** (key-free, runs anywhere, but lower-quality boundaries + fabricated server/shot metadata stays hidden) · **GPU WASB** (best quality, but needs a GPU host + is owner-parity-gated, not production-verified). *Recommend Gemini for "real-ish, cheapest" ? pending your key + OK (paid + uploads footage).*

36 3. **Engine host:** local Windows (simplest, Gemini/motion) · the CPU droplet `168.144.126.176` (shared testbed ? careful) · a GPU VM (only for WASB).

37 4. **Which bucket video:** e.g. `gs://khelsutra-rally-corpus/videos/mahadevpura\_smoke\_180s.mp4` (already staged, known-good) or another clean ?720p clip (must pass the engine validators).

38

39 ## 5. Honest caveats

40 - **Gemini = paid + uploads footage to Google** (an external/paid action ? needs your explicit OK + a clean clip). **Rotate the chat-shared `GEMINI\_API\_KEY`.

41 - **motion** fabricates `server`/`receiver`/`shot\_count` (the SPA correctly hides them) ? boundaries are real but metadata isn't.

42 - **GPU/native WASB** is partially wired + owner-byte-parity-gated ? **not production-verified**.

43 - Path A's "point at the bucket" is a **CLI pre-step**, not in-UI ? full in-UI ingest is Path B.

44 - Edge auth: none in the engine (localhost dry-run is fine; required before any non-localhost exposure).

45

46 ## 6. Definition of done (the dry run)

47 - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ? compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.

48 - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ? (Kannada also works via the same switcher.)

49 - [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST /api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p, 76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screencast:** `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.

50 - [] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ? droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a from-scratch engine install or a proper deploy of the fixed engine.

51

52 ## 7. ? Run result ? Path A, local (2026-06-21)

53 Ran the real CUJ end to end on the local Windows box, isolated engine off `origin/master` (fresh DB/output worktree), **Gemini** detector:

54 1. `gs://khelsutra-rally-corpus/videos/mahadevpura\_smoke\_180s.mp4` (1080p/30fps) ?
`gsutil cp` local ? `POST /api/process {segmenter: gemini}`.

55 2. Engine processed via Gemini (3 chunks) ? ****12 rallies**** (16 found, 4 too-short rejected), `video\_id 05e0e577?`.

56 3. SPA (`npm run dev`, Vite proxies `/api/?`:`8000`) opened at `?video=05e0e577?` ?
****loaded the 12 real rallies**** ("Loaded 12 rallies from this session").

57 4. ****Build ? /api/compile ? reel**** (1920x1080, 76.5s) ? ****downloaded to**
`~/Downloads/khelsutra-dryrun/highlights\_local\_en.mp4`****** (50.9 MB).

58 5. ****Repeated with the UI in Hindi**** (`?lang=hi`): the page + rally cards + "?? ????"
+ "???? ? ? ? ? ? ?" all in Hindi; reel compiled + ****downloaded**** (`highlights\_local\_hi.mp4`).
? multilingual workflow confirmed.

59

60 Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of
the i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.

61

62 **### ? Fix the dry run surfaced (engine) ? MERGED-PENDING**

63 The async-job server ****hung every job in `queued`**** when run the documented way (`python
-m backend.main`): main.py runs as `__main__` (sets `__main__.db_instance`), but `job_worker`
reads `import backend.main` ? a ****separate**** module whose `db\_instance` is `None` ? the drain
crashes silently. Latent because the unit tests monkeypatch the executor inline and the UI was
verified vs a stub engine. ****Fix + a real-subprocess regression test ? engine**
[#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281)****** (dispatch through
`backend.main`; FAILS-without / PASSES-with). Unblocked the run via the single-module
invocation; the fix applies to the droplet too.

64

65 **## 8. ? Correction (2026-06-21): B-P1 (the bucket-ingest endpoint) was already shipped ?**
verified against code

66 ****What this doc claimed (\$1, original):**** "`POST /api/process` accepts only a ****local****
`video\_path`" and "bridging `gs://` ? a queryable `video\_id` in the API DB is ****net-new****
(T12/T14)."

67

68 ****What the code actually does `[verified file:line, this session]`:**** `POST
/api/process` has accepted bucket/Drive URIs since ****engine M2**
([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280))******. A
`gs://`/`gs://`/`s3://`/`local://` `video\_path`:

69 1. is resolved at submit (`backend/main.py:274` ? `storage.resolve\_input\_ref`;
`gs://`/`gs` at `backend/storage/ref.py:60`; explicit scheme wins under default
`input\_backend=local` at `ref.py:87`),

70 2. ****skips the local-`os.path.exists` gate**** for non-local backends (`main.py:283`) ?
returns ****202, not 404****,

71 3. is fetched to scratch by the worker (`\_execute\_processing` ?
`storage.acquire\_local\_input`, `main.py:159` ? `GCSStorage.fetch\_to\_local`), then

72 4. runs the ****same**** pipeline as a local file, writing a queryable `video\_id` to the API
SQLite DB.

73

74 ****? Scope pin (default single-user mode):**** this holds with
`security.allowed\_video\_root` ****unset**** (the local-appliance default). In ****HOSTED mode****
(`allowed\_video\_root` set), a bucket URI is ****correctly rejected `400`**** ? a tenant can't point
the engine at an arbitrary bucket. That is by design, not a regression.

75

76 ****Already tested**** (so this can't silently regress):
`tests/test\_api\_endpoints.py::test\_process\_nonlocal\_uri\_skips\_exists\_and\_fetches`,
`?:test\_process\_hosted\_mode\_rejects\_nonlocal\_uri` (the hosted-mode counterweight),
`?:test\_process\_unknown\_scheme\_returns\_400`,
`?:test\_cli\_nonlocal\_input\_fetches\_and\_keeps\_uri`. So ****B-P1 needs zero new engine feature**
code** ? the only Path-B work is ****B-P2 (the SPA ingest/progress screen)****. *(The hermetic test
fakes the network download via `m.storage.fetch`, so it proves the CODE routing, not the live
GCS fetch ? that needs `google-cloud-storage` + ADC on the box, verified at deploy time / option
3.)*

77

78 ****Why it went stale (anti-spiral lesson):**** the \$1 claim was tagged `[verified]` but was
inferred from the ****worker CLI**** (`backend.worker infer`, which writes CSV to the bucket) rather
than checked against the ****API endpoint's**** routing code ? and M2/#280 had since wired URI
ingest into `/api/process`. ****The engine ships PRs between sessions, so its surface drifts.****
Convention going forward (also baked into the memory steering): ****before scoping any**

```

`[gap]`\`net-new` engine claim as work, re-verify it against engine code at a cited `file:line`
? a `[verified]` tag without a `file:line` is a hypothesis, not a fact.**
79
80     ## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)
81     The full Path-B flow, driven **entirely from the UI** (no CLI pre-step), against a real
local engine (`python -m backend.main --server`, Gemini detector,
`GOOGLE_APPLICATION_CREDENTIALS` for the GCS fetch):
82     1. Opened the SPA ? the **"Process a video" ingest panel (B-P2) ? pasted
`gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? **Process video**.
83     2. The engine **fetched the bucket video** (`google-cloud-storage` + ADC) ? **Gemini**
segmented it (live "Processing? segmenting (gemini)" progress in the UI) ? **12 rallies**
(`video_id 05e0e577?`), handed off via `?video=` ? the existing back-half rendered them.
84     3. **Build reel** ? `/api/compile` ? **reel preview + download** ? MP4 saved locally
(**1920x1080, 76s**).
85     4. **Screencast of the whole CUJ** (Playwright-recorded; the ~3-min Gemini wait fast-
forwarded): `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
86
87     ### ? Bug the run surfaced + fixed (engine) ? MERGED
88     `/api/compile` **500'd** on a bucket-ingested video: `videos.filepath` keeps the
**source URI** (`gs://`, M2 provenance), and the stitcher was handed that directly ? ffmpeg
can't open a `gs://` path. **Fix:** compile now resolves the source via
`storage.acquire_local_input` first (identity for a local path; fetch-to-scratch for a bucket
URI) + a regression test ? engine **[#289](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/289)** (full suite 1182?). This is what makes the in-UI bucket?reel CUJ actually
complete. *Proof-and-validation-driven: recording the screencast is what found the bug.*
89
90     ### Changelog
91     - `2026-06-21` ? created; Path-A run recorded ($7).
92     - `2026-06-21` ? **$8 correction:** verified B-P1 (`gs://`?`/api/process`?API-DB
`video_id`) is already shipped + tested (M2/#280); $1/$2/$4/$6 reframed so Path B = SPA-only
(B-P2). Sibling stale claims fixed in `LOCAL_APPLIANCE_ARCHITECTURE.md` (`/api/health` exists;
bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.
93     - `2026-06-21` ? **$9: Path-B live e2e DONE** ? the in-UI `gs://`?reel CUJ ran end to
end (12 rallies ? reel downloaded); surfaced + fixed the compile-from-bucket 500 (engine #289);
screencast uploaded to the promo bucket. **DoD $6 box 3 ?.**
94

```

4. user (2026-06-22T09:32:50.201Z)

```

50b2bd7 chore(repo): ignore local runtime artifacts + correct stale NEXT_STEPS claims (#94)
e2570e4 docs: archive the Friendly & Multilingual tracker (engineering complete) (#93)
0bef61e docs(friendly-multilingual): close out the chapter ? log #89/#90/#91 + strengthen fitNav
tests (#92)
9f77ddc test(site): browser layout-regression CI guard + auto-collapse the nav to fit every
locale (#91)
4415fe8 feat(site): localize remaining home sections (6 langs) + smooth the responsive header
(#90)
26c9da1 feat(site): localize the hero demo mockup + #how feature block (6 languages) (#89)
7ae9cee feat(web): P2c first-run setup wizard (3 steps from /api/capabilities, 6 languages)
(#88)
cf59fbf feat(site): add es/da/id to the marketing site (parity with the SPA's 6 languages) (#87)
17242a9 feat(site): localize own-model body content to kn/hi (#86)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
5e5a2ac docs(friendly-multilingual): close out the chapter ($9 DoD + UI/UX + B7 deferred) (#84)
4c0f35c fix(site): UI/UX ? compact language switcher + header overflow fix + localized hero
(#83)
0fae364 feat(web+site): A2 ? localized WhatsApp share for the reel (SPA + site) (#82)
1e52cfd deploy(droplet): tailored alpha kit for the CPU droplet (engine api. side) (#81)
64076a5 test(web): add a coverage floor (CI-gated) + i18n regression tests for the #77 audit
(#80)
8328721 docs(tracker): refresh the $7 milestone prose to merged reality (A6/A7/#77 landed) (#79)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
9f1ec9c fix(i18n): native-review pass on the kn/hi SPA console catalogs (#77)

```

9a2d7d2 feat(site): A7 ? camera-placement diagram on the capture page (#76)
aa8d5e9 feat(site): A6 ? plain-language Request-Access GPU/storage questions (#75)

5. assistant (2026-06-22T09:32:53.443Z)

This is a long, detailed doc. Let me verify the key falsifiable claims. I'll check the engine repo for the cited file:line claims, the SPA front-half existence, PR numbers, and the test names. Note this doc references the engine repo (`badminton-highlight-indexer`) heavily.

6. user (2026-06-22T09:32:55.544Z)

docs/DRY_RUN_CUJ.md
docs/LOCAL_APPLIANCE_ARCHITECTURE.md
docs/PER_RALLY_SELECTION.md

7. user (2026-06-22T09:32:57.258Z)

NEXT_STEPS.md
README.md
deploy
docs
locales
node_modules
site
web
---SITE---
README.md
access.db
make_og.py
node_modules
package-lock.json
package.json
public
schema.sql
scripts
server.js
src
wrangler.toml
---WEB---
README.md
dist
index.html
node_modules
package-lock.json
package.json
src
tsconfig.json
vite.config.ts
---SPA SRC---
web/src/components/CompileResult.tsx
web/src/components/IngestPanel.test.tsx
web/src/components/IngestPanel.tsx
web/src/components/QueryBar.examples.test.ts
web/src/components/QueryBar.test.tsx
web/src/components/QueryBar.tsx
web/src/hooks/useIngestJob.test.ts
web/src/hooks/useIngestJob.ts
web/src/lib/query.test.ts
web/src/lib/query.ts

8. assistant (2026-06-22T09:33:00.897Z)

The SPA front-half (`IngestPanel.tsx`, `useIngestJob.ts`) now exists in the khelsutra repo's `web/` directory. This is important ? the doc describes B-P2 as work that was done. Let me

examine the SPA ingest panel and the engine repo file:line claims.

9. user (2026-06-22T09:33:01.878Z)

```
1      // The front-half ingest screen (appliance S1/S2; tracker B-P2 + T4). Two ways to point
the tool at a
2      // video: UPLOAD a local file (primary ? alpha users have a file, not a URI; chunked to
the engine's
3      // /api/upload/* endpoints) or paste a `gs://` bucket URI / host path (secondary). Both
converge on
4      // the SAME back-half: POST /api/process ? poll GET /api/jobs/{id} ? hand the video_id
to App. Honest
5      // by construction: the engine reports coarse phase WORDS (not a %), and a success-
shaped failure
6      // (done but no rallies) is shown, never silently treated as a load.
7      import { useState } from "react";
8      import type { ChangeEvent, FormEvent } from "react";
9      import { useTranslation } from "react-i18next";
10     import { useIngestJob } from "../hooks/useIngestJob";
11     import { useUpload } from "../hooks/useUpload";
12     import { errorText } from "../lib/errorText";
13     import { StatusChip, Spinner } from "../StatusChip";
14
15     interface Props {
16       /** Called with the engine's video_id once a submitted job finishes successfully. */
17       onLoaded: (videoId: string) => void;
18     }
19
20     export function IngestPanel({ onLoaded }: Props) {
21       const { t, i18n } = useTranslation();
22       const [uri, setUri] = useState("");
23       const { phase, submit, cancel: cancelIngest, reset: resetIngest, busy: ingestBusy } =
useIngestJob(onLoaded);
24       // The UI locale recorded with the submission (PMF analytics + the localized reel
watermark).
25       const locale = i18n.resolvedLanguage ?? i18n.language;
26       // On upload completion, feed the returned server PATH straight into the engine
processing back-half.
27       const upload = useUpload((path) => submit(path, locale));
28       const busy = ingestBusy || upload.busy;
29
30       const onSubmitUri = (e: FormEvent) => {
31         e.preventDefault();
32         submit(uri, locale);
33       };
34       const onFile = (e: ChangeEvent<HTMLInputElement>) => {
35         const file = e.target.files?.[0];
36         if (file) upload.start(file);
37         e.target.value = ""; // let re-picking the same file fire change again
38       };
39       const cancelAll = () => {
40         upload.cancel();
41         cancelIngest();
42       };
43       const resetAll = () => {
44         upload.reset();
45         resetIngest();
46       };
47
48       // Shared mapping with the build flow (lib/errorText): an `engine` failure carries the
engine's own
49       // message (a 400 scheme/path/extension detail, a worker error); every other code maps
to a friendly
50       // localized line. Surface whichever half failed (upload or engine) ? they're mutually
exclusive.
```

```

51     const errorState = upload.phase.status === "error" ? upload.phase.error : phase.status
=== "error" ? phase.error : null;
52     const errorMsg = errorState ? errorText(t, errorState) : null;
53
54     return (
55         <section className="mt-6 rounded-2xl border border-black/10 bg-white p-5" data-
testid="ingest-panel">
56             <h2 className="text-lg font-bold">{t("ingest.title")}</h2>
57             <p className="mt-1 max-w-2xl text-sm text-ink/60">{t("ingest.intro")}</p>
58
59             {/* PRIMARY: upload a local file (T4). A native file input, visually a button
(?44px, A4). */}
60             <div className="mt-4">
61                 <label
62                     className={
63                         // focus-within makes the visible button show a focus ring when the (sr-
only) input is
64                         // focused ? a keyboard user needs a visible focus indicator (WCAG 2.4.7).
65                         "inline-flex min-h-11 items-center rounded-xl bg-ink px-4 py-2 text-sm font-
semibold text-white hover:bg-ink2 focus-within:outline-none focus-within:ring-2 focus-
within:ring-green focus-within:ring-offset-2 " +
66                         (busy ? "pointer-events-none opacity-40" : "cursor-pointer")
67                     }
68                 >
69                     {t("ingest.chooseFile")}
70                     <input
71                         type="file"
72                         accept=".mp4,.mov,video/mp4,video/quicktime"
73                         aria-label={t("ingest.uploadAria")}
74                         disabled={busy}
75                         onChange={onFile}
76                         className="sr-only"
77                     />
78                 </label>
79                 <p className="mt-2 text-xs text-ink/40">{t("ingest.fileHint")}</p>
80             </div>
81
82             {/* SECONDARY: or point at a cloud bucket / host path. */}
83             <div className="mt-4 border-t border-black/5 pt-4">
84                 <p className="text-xs font-semibold uppercase tracking-wide text-
ink/40">{t("ingest.orLink")}</p>
85                 <form onSubmit={onSubmitUri} className="mt-2 flex flex-wrap items-center gap-2">
86                     <input
87                         type="text"
88                         value={uri}
89                         onChange={(e) => setUri(e.target.value)}
90                         placeholder={t("ingest.uriPlaceholder")}
91                         aria-label={t("ingest.uriAriaLabel")}
92                         disabled={busy}
93                         className="min-h-11 min-w-[18rem] flex-1 rounded-xl border border-black/15
bg-white px-4 py-2 text-sm text-ink outline-none focus:border-green disabled:opacity-50"
94                     />
95                     <button
96                         type="submit"
97                         disabled={busy}
98                         className="min-h-11 rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-
white disabled:opacity-40"
99                     >
100                         {phase.status === "submitting" ? t("ingest.submitting") :
t("ingest.submit")}
101                     </button>
102                 </form>
103                 <p className="mt-2 text-xs text-ink/40">{t("ingest.hint")}</p>
104             </div>
105

```



```

106      { /* Live status (aria-live so screen readers hear progress) as icon+text chips ?
never icon-only
107      (A8). Upload shows a determinate % bar; the engine phase is indeterminate
(phase words). */}
108      <div aria-live="polite" className="mt-3 text-sm" data-testid="ingest-status">
109          {upload.phase.status === "uploading" && (
110              <div data-testid="upload-progress">
111                  { /* Static chip label (not the per-tick %) so the aria-live region announces
once; the live
112                      % rides the progressbar's aria-valuenow + a sighted-only label (A8/T4).
*/}
113                      <StatusChip tone="info" icon={<Spinner />}
label={t("ingest.uploadingStatus")} />
114                      <div className="mt-2 flex max-w-sm items-center gap-2">
115                          <div
116                              className="h-1.5 flex-1 overflow-hidden rounded-full bg-black/10"
117                              role="progressbar"
118                              aria-label={t("ingest.uploadProgress")}
119                              aria-valuenow={upload.phase.pct}
120                              aria-valuemin={0}
121                              aria-valuemax={100}
122                          >
123                              <div className="h-full bg-green transition-all" style={{ width:
`${upload.phase.pct}%` }} />
124                          </div>
125                          <span aria-hidden="true" className="text-xs tabular-nums text-ink/50">
126                              {upload.phase.pct}%
127                          </span>
128                      </div>
129                      <p className="mt-2 text-xs text-ink/50">{t("ingest.reassure")}</p>
130                      <button
131                          type="button"
132                          onClick={cancelAll}
133                          className="mt-2 inline-flex min-h-11 items-center rounded-xl border
border-black/15 px-4 text-sm hover:border-green"
134                      >
135                          {t("ingest.cancel")}
136                      </button>
137                  </div>
138              )}
139          {ingestBusy && (
140              <div>
141                  <StatusChip
142                      tone="info"
143                      icon={<Spinner />}
144                      label={
145                          t("ingest.working") +
146                          (phase.status === "polling" && phase.progress
147                              ? ` . ${t("ingest.progress", { progress: phase.progress })}`
148                              : "")
149                      }
150                  />
151                  { /* Reassurance for slow uplinks: the job runs on the server, so it's safe
to wait. */}
152                  <p className="mt-2 text-xs text-ink/50">{t("ingest.reassure")}</p>
153                  <button
154                      type="button"
155                      onClick={cancelAll}
156                      className="mt-2 inline-flex min-h-11 items-center rounded-xl border
border-black/15 px-4 text-sm hover:border-green"
157                  >
158                      {t("ingest.cancel")}
159                  </button>
160              </div>
161          )}

```

```

162         {errorMsg && (
163             <div className="rounded-lg border border-red-200 bg-red-50 px-3 py-2 text-
red-800" data-testid="ingest-error">
164                 <StatusChip tone="error" icon="?" label={t("ingest.statusFailed")} />
165                 <p className="mt-2">{errorMsg}</p>
166                 <button
167                     type="button"
168                     onClick={resetAll}
169                     className="mt-2 inline-flex min-h-11 items-center rounded-lg border
border-red-300 px-3 text-sm hover:bg-red-100"
170                 >
171                     {t("ingest.retry")}
172                 </button>
173             </div>
174         )}
175     </div>
176 </section>
177 );
178 }
179

```

10. user (2026-06-22T09:33:02.064Z)

```

1      // Drives the front-half ingest: submit a video to POST /api/process, then poll GET
/api/jobs/{id}
2      // until it finishes, and hand the resulting video_id to the existing rally-loading
back-half.
3      //
4      // Kept i18n-FREE (returns stable IngestErrorCodes, not prose) so it unit-tests without
a
5      // translation context ? IngestPanel maps codes to localized strings. The poll loop is a
RECURSIVE
6      // setTimeout (never setInterval): each tick waits for the prior getJob to resolve, so a
slow engine
7      // can't pile up overlapping requests. It backs off, gives up after a ceiling, and is
fully
8      // cancellable on unmount / user-cancel / a superseding submit.
9      import { useCallback, useEffect, useRef, useState } from "react";
10     import { getJob, processVideo } from "../api/client";
11     import { classifyError, type IngestError } from "../lib/errors";
12     import { log, newCorrelationId } from "../lib/log";
13     import type { JobStatus } from "../api/types";
14
15     const POLL_MIN_MS = 1000;
16     const POLL_MAX_MS = 5000;
17     const POLL_BACKOFF = 1.5;
18     const POLL_CEILING_MS = 10 * 60 * 1000; // a wedged 'running' job won't poll forever
19
20     export type IngestPhase =
21     | { status: "idle" }
22     | { status: "submitting" }
23     | { status: "polling"; progress: string | null }
24     | { status: "done"; videoId: string }
25     | { status: "error"; error: IngestError };
26
27     interface Run {
28         cancelled: boolean;
29         timer?: ReturnType<typeof setTimeout>;
30     }
31
32     export function useIngestJob(onLoaded?: (videoId: string) => void) {
33         const [phase, setPhase] = useState<IngestPhase>({ status: "idle" });
34         const runRef = useRef<Run | null>(null);
35         // Keep the latest onLoaded without making submit depend on it (a parent re-render
won't restart polls).

```

```

36     const onLoadedRef = useRef(onLoaded);
37     onLoadedRef.current = onLoaded;
38
39     const stop = useCallback(() => {
40         const run = runRef.current;
41         if (run) {
42             run.cancelled = true;
43             if (run.timer) clearTimeout(run.timer);
44         }
45         runRef.current = null;
46     }, []);
47
48     // Cancel any in-flight poll loop when the component unmounts (App.tsx:80-82 alive-
49     // flag idiom).
50     useEffect(() => stop, [stop]);
51
52     const submit = useCallback(
53         (rawUri: string, locale?: string) => {
54             const uri = rawUri.trim();
55             if (!uri) {
56                 setPhase({ status: "error", error: { code: "invalidUri" } });
57                 return;
58             }
59             stop(); // supersede any prior run
60             const run: Run = { cancelled: false };
61             runRef.current = run;
62
63             const cid = newCorrelationId();
64             const startedAt = Date.now();
65             setPhase({ status: "submitting" });
66             log("info", "ingest.submit_start", { cid, uri });
67
68             const fail = (error: IngestError, event: string) => {
69                 if (run.cancelled) return;
70                 setPhase({ status: "error", error });
71                 const level = error.code === "timeout" || error.code === "emptyResult" ? "warn"
72                 : "error";
73                 log(level, event, { cid, uri, code: error.code, detail: error.detail });
74             };
75
76             const poll = (jobId: string, delay: number) => {
77                 run.timer = setTimeout(() => {
78                     void (async () => {
79                         if (run.cancelled) return;
80                         let job: JobStatus;
81                         try {
82                             job = await getJob(jobId, cid);
83                         } catch (err) {
84                             fail(classifyError(err), "ingest.poll_error");
85                             return;
86                         }
87                         if (run.cancelled) return;
88
89                         if (job.status === "done") {
90                             const videoId = job.video_id ?? job.result?.video_id ?? null;
91                             // Two status layers: a `done` job with a `failed` pipeline verdict (or no
92                             // video_id) is
93                             // a success-shaped FAILURE ? never load it, never hang.
94                             if (job.result?.status === "failed" || !videoId) {
95                                 fail({ code: "emptyResult" }, "ingest.empty_result");
96                                 return;
97                             }
98                             setPhase({ status: "done", videoId });
99                             log("info", "ingest.done", { cid, uri, video_id: videoId });

```

```

98         onLoadedRef.current?.(videoId);
99         return;
100     }
101     if (job.status === "failed") {
102         fail({ code: "engine", detail: job.error ?? "job failed" },
"ingest.job_failed");
103         return;
104     }
105     // queued | running ? reflect coarse progress (free-text phase) and keep
polling.
106     setPhase({ status: "polling", progress: job.progress ?? null });
107     if (Date.now() - startedAt > POLL_CEILING_MS) {
108         fail({ code: "timeout" }, "ingest.poll_timeout");
109         return;
110     }
111     poll(jobId, Math.min(delay * POLL_BACKOFF, POLL_MAX_MS));
112     })();
113     }, delay);
114 };
115
116 void (async () => {
117     let jobId: string;
118     try {
119         const resp = await processVideo(uri, cid, locale);
120         jobId = resp.job_id;
121     } catch (err) {
122         fail(classifyError(err), "ingest.submit_error");
123         return;
124     }
125     if (run.cancelled) return;
126     log("info", "ingest.queued", { cid, uri, job_id: jobId });
127     setPhase({ status: "polling", progress: null });
128     poll(jobId, POLL_MIN_MS);
129     })();
130 },
131 [stop],
132 );
133
134 const cancel = useCallback(() => {
135     stop();
136     setPhase({ status: "idle" });
137     log("info", "ingest.cancel", {});
138 }, [stop]);
139
140 const reset = useCallback(() => {
141     stop();
142     setPhase({ status: "idle" });
143 }, [stop]);
144
145 const busy = phase.status === "submitting" || phase.status === "polling";
146 return { phase, submit, cancel, reset, busy };
147 }
148

```

11. assistant (2026-06-22T09:33:06.487Z)

This is significant. The doc (DRAFT, dated 2026-06-21) describes B-P2 (the SPA ingest/process screen) as the remaining net-new work and §6 box 3 as done via a "live e2e" run. But the SPA ingest panel is now fully built and merged into the `web/` directory (with upload support too ? T4). Let me confirm when `IngestPanel.tsx` landed and check the git history for B-P2.

12. user (2026-06-22T09:33:09.929Z)

=== IngestPanel history ===

c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process

```
(analytics) (#73)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
(#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
=== useIngestJob history ===
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
=== first commit dates ===
2026-06-21 15:45:54 +0530 4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video
? process ? load) (#56)
2026-06-21 20:35:27 +0530 95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip
text (#61)
2026-06-21 20:52:13 +0530 501c014 feat(web): A5 ? friendly engine-error mapping across build +
load (shared errors.*) (#62)
```

13. user (2026-06-22T09:33:10.623Z)

```
=== B-P2 / ingest / process mentions in commit log ===
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download,
multilingual) (#53)
```

14. assistant (2026-06-22T09:33:15.664Z)

B-P2 (#56) landed 2026-06-21 ? consistent with the doc's §9 claim. The doc itself is internally accurate about the SPA work shipping. Since then, T4 (in-UI file upload, #66) was added on top, which the doc doesn't mention but isn't a contradiction.

Now let me verify the engine repo file:line claims (`backend/main.py:274,283`, `backend/storage/ref.py:60,87`, `main.py:159`, test names) and the engine PR numbers (#280, #281, #289). These reference the sibling engine repo.

15. user (2026-06-22T09:33:19.526Z)

```
=== branch/head ===
work/golden-corpus-hygiene
689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)
=== files exist? ===
backend/main.py
backend/storage/ref.py
=== ref.py line 60 ===
    m = _SCHEME_RE.match(s)
    if not m:
        return StorageRef("local", "", s, f"local://{s}")
    scheme = m.group(1).lower()
    rest = s[m.end():]
    if scheme == "gs":
        scheme = "gcs"
    if scheme in ("local", "gdrive"):
        # Path-keyed, no bucket: local FS, or a path under the mounted Google Drive "My Drive".
        return StorageRef(scheme, "", rest, s)
    if scheme in _KNOWN:
        bucket, _, key = rest.partition("/")
        return StorageRef(scheme, bucket, key, s)
    raise ValueError(f"unsupported storage URI scheme: {scheme}:// (in {s!r})")
```

```
def resolve_input_uri(raw: str, *, input_backend: str = "local", input_root: str = "") -> str:
    """Resolve a user-supplied input reference into a storage URI (COMPUTE_DECOUPLED_SERVING
M2).

    Precedence (first match wins):
    1. An explicit ``scheme://`` on ``raw`` (``gs://``, ``s3://``, ``local://``,
``gdrive://``) ? as-is.
    2. An ABSOLUTE local path (``~/data/x.mp4``, ``C:\\?``, ``C:/?``) ? as-is (always local).
    3. A bare/relative name + a non-empty ``input_root`` ? joined under it
    (``gs://bucket/videos`` + ``x.mp4`` ? ``gs://bucket/videos/x.mp4``).
    4. A bare/relative name + ``input_backend != "local"`` ? prefixed (``gcs`` +
``bucket/x`` ?
    ``gcs://bucket/x``).
    5. Otherwise ? ``raw`` unchanged (a relative/bare LOCAL path ? today's behaviour).

    So the default (``input_backend="local"`` , ``input_root=""``) is an identity passthrough for
every
    path used today, preserving byte-for-byte behaviour.
    """
    s = str(raw)
    if _SCHEME_RE.match(s):          # 1. explicit scheme wins
        return s
    # 2. an anchored LOCAL path is always local, never joined. Detect this OS-INDEPENDENTLY ?
    # os.path.isabs is platform- AND Python-version-dependent (on Windows + py3.13 a POSIX-
style
    # "/data/x" is NOT absolute). Anchored == a leading "/" or "\\ ", or a Windows drive
prefix "X:".
    if s.startswith((" /", "\\ ")) or (len(s) >= 2 and s[1] == ":" and s[0].isalpha()):
```

16. assistant (2026-06-22T09:33:25.944Z)

Important: I'm in the engine repo but on branch `work/golden-corpus-hygiene`, not master. The doc's file:line claims are about the engine, but the engine repo here is the parent session's cwd, not the khelsutra repo I'm auditing. Note: the audit instructions say verify against the khelsutra repo at master HEAD (#94), and the engine claims are cross-repo. Let me confirm the `gs://`?`gcs` normalization is at line 60 (it is ? `if scheme == "gs": scheme = "gcs"` is at line 60). The explicit-scheme-wins is at the resolve function. The doc says "ref.py:87" for explicit scheme ? let me note the doc's line numbers may be approximate/drifted since the engine moved.

Let me verify the engine PRs and test names exist, and the `backend/main.py` line claims.

17. user (2026-06-22T09:33:27.218Z)

=== main.py around 159, 274, 283 ===

```
def _open_browser_when_ready(url: str, health_url: str, timeout: float = 60.0) -> bool:
    """Poll `health_url` until the server is up, then open `url` in the user's default browser.
    Returns True if a browser open was attempted. No-op (returns False) when RALLY_NO_BROWSER is
set
    (tests / headless) or the server never becomes healthy (don't open a dead page). Never
raises."""
    if os.environ.get("RALLY_NO_BROWSER"):
        return False
    server: Optional[str] = None
    receiver: Optional[str] = None
    duration_min: Optional[float] = None
    event_type: Optional[str] = None

class CompileRequest(BaseModel):
    video_id: str
    segment_ids: List[int]
    output_filename: str
    # The UI locale the SPA was rendered in (e.g. "kn", "hi-IN"). Localizes the reel watermark
```

```

# (text + script font). Optional ? absent/unknown keeps the configured default (en
behavior).
    locale: Optional[str] = None

def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
    """Commit a (re)segmentation with destroy-AFTER-confirm semantics.

    On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
    fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark) and
    keep the prior good results intact ? a failed/empty re-run never destroys prior data.
    """
    if segments:

```

18. user (2026-06-22T09:33:30.918Z)

```

=== engine PRs 280/281/289 in log ===
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching
(#289)
7bbdd19 fix(api): resolve a bucket-ingested video's source to a local file before stitching
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
a382cf8 docs(serving): mark $7 tracker M2 ? + changelog (PR #280)
6554719 feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
fb51cdc docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($) (#267)
8d8d230 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; D0=train,
GCP=serve (ADR-011) (#260)
4b2c202 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; D0=train,
GCP=serve (ADR-011)
d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock
(#254)
6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)
6878d45 feat(storage): pluggable StorageBackend ? local + S3/GCS first-class buckets (#249)
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
=== test names ===
tests/test_api_endpoints.py:508:def test_process_nonlocal_uri_skips_exists_and_fetches(env):
tests/test_api_endpoints.py:541:     Counterweight =
test_process_hosted_mode_rejects_nonlocal_uri (gs:// -> 400 when allowed_video_root
tests/test_api_endpoints.py:577:def test_process_unknown_scheme_returns_400(env):
tests/test_api_endpoints.py:608:def test_process_hosted_mode_rejects_nonlocal_uri(env):
tests/test_api_endpoints.py:627:def test_cli_nonlocal_input_fetches_and_keeps_uri(tmp_path,
monkeypatch):
Binary file tests/__pycache__/test_api_endpoints.cpython-313-pytest-9.0.3.pyc matches

```

19. assistant (2026-06-22T09:33:35.650Z)

All four cited engine tests exist; engine PRs #280 (M2), #281 (job-drain fix), #289 (compile-from-bucket fix) all exist with descriptions matching the doc. The doc's engine claims are well-grounded. The line numbers (159, 274, 283) have drifted slightly since the engine moved past those commits (the file content at those exact lines no longer matches the cited descriptions), but that's expected drift in a cross-repo claim ? the substance holds. Let me verify the substance of the cited lines via grep rather than absolute line numbers.

20. user (2026-06-22T09:33:37.299Z)

```

=== resolve_input_ref + exists skip ===
210:     if os.path.exists(index_path):
366:         "wasb_weights_present": bool(weights_path) and os.path.exists(weights_path),
458:     local_video = storage.acquire_local_input(req.video_path, getattr(global_config,
"storage", None))
581:     input_ref = storage.resolve_input_ref(req.video_path, getattr(global_config,

```

```

"storage", None))
586:     # bucket/Drive URI can't be cheaply os.path.exists()'d ? its existence is verified when
the worker
589:     # rejects a non-local URI as out-of-root.
590:     if input_ref.backend == "local" and not os.path.exists(input_ref.key):
764:         local_src = storage.acquire_local_input(video["filepath"], getattr(global_config,
"storage", None))
1004:         input_ref = storage.resolve_input_ref(args.video_path, storage_cfg)
1008:         if input_ref.backend == "local" and not os.path.exists(input_ref.key):
1011:         local_video = storage.acquire_local_input(args.video_path, storage_cfg)

```

21. user (2026-06-22T09:33:39.929Z)

=== PER_RALLY_SELECTION exists / status ===

Interactive per-rally selection ? reel builder (RallyPicker)

```

> **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ? select ?
compile ? download, on existing engine endpoints). **Validated live e2e against a REAL engine
(Gemini) via the in-UI bucket?reel CUJ ? `DRY_RUN_CUJ.md` §9, 2026-06-21** (supersedes the
original stub-engine verification). Design **APPROVED by owner 2026-06-20**. Downstream
milestones M1?M5 (multi-video auth, per-rally preview endpoint, corpus loop, localized NL)
remain owner-gated. . **Owner:** Avi . **Created:** 2026-06-20 . **Last updated:** 2026-06-21
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`;
pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
> **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
> **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine code
this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.

```

0. TL;DR

Build the ****pick-rallies ? reel**** surface as a ****hybrid****: a selectable rally ****list/grid**** is the workhorse; a ****deterministic query bar**** ("rallies over 10s, longest first") pre-selects into the ****same**** selection set; a sticky footer compiles the selection into one downloadable MP4. The entire MVP runs on ****existing engine endpoints**** ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396, 307-328]` ? with ****zero new engine intelligence and zero new endpoints****. The single most important caveat: queries and chips are scoped to ****length / count / order / shot-count only****; shot-type, player, and score are ****roadmap-only**** (no detector, no column) and must not be faked. The biggest UX-vs-effort tension: ****no per-rally preview is possible today**** ? only the ****final compiled reel**** can stream ? so MVP selects on text metadata, and visual preview/timeline is a deliberate later milestone.

1. Problem & goal

Today the alpha can detect rallies and stitch a reel, but reel-building is either a ****silent `stitching.min\_shot\_count` threshold**** `[verified backend/main.py:362-383]` or a static checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies` while the API returns `play\_segments`, so it renders nothing `[verified backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is ****dynamic per-video selection, not a fixed threshold****: a coach should say "give me the long rallies from this session" and get exactly those, then fine-tune by hand.

****The concrete outcome ("good")**** A coach uploads a 40-minute session; the engine finds 60 rallies. They type ****"top 15 longest"**, eyeball the auto-selected cards, untick two warm-up rallies, name it ****"Tuesday ? long points"**, and download a ~6-minute reel ? without learning any threshold knob.****

****Goal:**** the smallest ***honest*** surface that turns engine-detected rallies into a user-curated, downloadable reel, with both ****fast**** (query) and ****precise**** (checkbox) selection sharing ****one**** state.

```
=== api client process/getJob ===
web/src/api/client.test.ts:44:   expect(highlightUrl("abc123", "my
reel.mp4")).toBe("/api/highlights/abc123/my%20reel.mp4");
web/src/api/client.test.ts:49:   it("POSTs video_id to /api/query and returns play_segments",
async () => {
web/src/api/client.test.ts:59:     expect(url).toBe("/api/query");
web/src/api/client.test.ts:72:   it("POSTs the selection to /api/compile", async () => {
web/src/api/client.test.ts:80:     expect(url).toBe("/api/compile");
web/src/api/client.ts:83: *   processing. POST /api/process {video_path} ? 202 {job_id, ...}
[main.py:260]. Fast-fails 400
web/src/api/client.ts:85:export async function processVideo(videoPath: string, cid?: string,
locale?: string): Promise<ProcessResponse> {
web/src/api/client.ts:95:/** Poll a submitted job. GET /api/jobs/{job_id} [main.py:303]. 404 ?
unknown job_id. */
web/src/api/client.ts:96:export async function getJob(jobId: string, cid?: string):
Promise<JobStatus> {
web/src/api/client.ts:100:/** List a video's detected rallies. POST /api/query {video_id} ?
{play_segments} [main.py:331]. */
```

22. assistant (2026-06-22T09:33:49.675Z)

The substance is all verified: ``resolve_input_ref`` + the non-local-skips-exists gate (``main.py:590``: ``if input_ref.backend == "local" and not os.path.exists(...)``) exactly matches the doc's claim. The SPA api client has ``processVideo`` ? ``/api/process`` and ``getJob`` ? ``/api/jobs``. The back-half routes (``/api/query``, ``/api/compile``, ``/api/highlights``) all exist.

Now let me verify the multilingual claim. The doc says en/kn/hi (3 languages) for the SPA. But the git log shows es/da/id were added (#87) bringing the SPA to 6 languages. The doc was written when it was 3. Let me confirm the current language set, and check the \$6 box 4 (CPU droplet) claim which is the one open item.

23. user (2026-06-22T09:33:52.313Z)

```
=== SPA locales ===
catalog-invariants.test.ts
hygiene.test.ts
i18n.test.tsx
index.ts
web/src/api/locale.test.ts
web/src/i18n/catalog-invariants.test.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
```

24. user (2026-06-22T09:33:54.818Z)

```
=== archives dir ===
FRIENDLY_AND_MULTILINGUAL.md
=== does DRY_RUN appear in README/NEXT_STEPS ===
docs/README.md:4:- **`DRY_RUN_CUJ.md`** ? ? **DELIVERED (2026-06-21):** the UI drives the tool
on a **GCS-bucket
```

25. user (2026-06-22T09:34:00.481Z)

```
=== SUPPORTED langs in i18n index ===
1:// i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
3:// Per docs/archives/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4://   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5://   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in the
catalog,
```

```

6://      not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
8://      - always-fallback to `en` so a missing translation degrades to English, never a raw key.
13:// Catalogs are HOISTED to repo-root /locales (D-CATALOG-HOME) ? the single source of truth
shared
14:// by this SPA and the marketing site's t() shim (tracker B3). Vite resolves the workspace
root via
15:// the repo `.git` marker, so importing the parent-dir /locales works in dev, build, and
viteest.
16:import enCommon from "../../locales/en/common.json";
17:import knCommon from "../../locales/kn/common.json";
18:import hiCommon from "../../locales/hi/common.json";
19:import esCommon from "../../locales/es/common.json";
20:import daCommon from "../../locales/da/common.json";
21:import idCommon from "../../locales/id/common.json";
23:// Supported locales: en/kn/hi foundation (FRIENDLY D6) + MULTILINGUAL_EXPANSION Phase A
(es/da/id,
24:// Latin, catalog-only). Anything else falls back to `en`.
25:export const SUPPORTED_LOCALES = ["en", "kn", "hi", "es", "da", "id"] as const;
27:export const DEFAULT_LOCALE: Locale = "en";
29:// ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker B3),
so a
30:// visitor's language choice carries across both surfaces.
33:// Catalogs live in repo-root /locales (D-CATALOG-HOME), consumed by both surfaces. `kn` +
`hi`
34:// shipped (B5/B6, AI-drafted ? native review pending). A missing key in any locale still
falls back
35:// to `en` ? the by-design invariant.
36:const resources = {
37:  en: { common: enCommon },
38:  kn: { common: knCommon },
39:  hi: { common: hiCommon },
40:  es: { common: esCommon },
41:  da: { common: daCommon },

```

26. user (2026-06-22T09:34:01.033Z)

```

1      # khelsutra docs index
2
3      ## Tracked work (this repo)
4      - **`DRY_RUN_CUJ.md`** ? ? **DELIVERED (2026-06-21):** the UI drives the tool on a
**GCS-bucket
5      video** ? review/select ? compile ? **reel downloaded local**, verified in en + Hindi.
Path A
6      (local) + Path B/B-P2 (in-UI `gs://` ingest, [#56]) both live e2e with a screencast
($9); the
7      only remaining box (droplet, option 3) is **parked by the owner** (GPU subscription).
Two engine
8      bugs the runs surfaced are fixed (#281, #289). Was a milestone of
`LOCAL_APPLIANCE_ARCHITECTURE.md`.
9      - **`archives/FRIENDLY_AND_MULTILINGUAL.md`** ? ? **DONE ? ARCHIVED (2026-06-22).** Made
the product
10     super user-friendly for non-tech-savvy coaches/parents **and** localized it to all **6
languages**
11     across both surfaces (`web/` SPA ? react-i18next, `site/` ? a shared `t()` shim); the
home-page body
12     is fully localized (except the deferred `#join` form, A6) and the responsive header is
fixed +

```

27. assistant (2026-06-22T09:34:09.557Z)

Now I have a clear picture. The SPA now supports 6 languages (en/kn/hi/es/da/id), but the DRY_RUN_CUJ.md doc repeatedly says the SPA/multilingual workflow is "en/kn/hi" (\$1, \$7). That's stale ? es/da/id were added in #87. However, the doc's CORE claim (multilingual workflow works, proven in Hindi + Kannada) remains true; it's just that the language count is now larger.

Let me verify the \$6 box 4 droplet claim ("parked/un-provisioned") is still the open item, and check the `docs/README.md` index status which says "DELIVERED". The doc itself is still marked `DRAFT` in its header ? that's a status mismatch worth flagging. Let me confirm the README's "parked by the owner" framing vs the doc's "un-provisioned" framing, and double-check there's nothing claiming the doc is archived.

The doc header says `Status: DRAFT ? owner-gated · Goal for this session`. But the work is DONE (DoD \$6 boxes 1-3 checked, box 4 parked) and `docs/README.md` calls it "DELIVERED". The DRAFT status is stale.

Let me do a final check on the `docs/README.md` reference to #56 and confirm the doc is not yet archived (it lives at top-level `docs/`, not `docs/archives/`).

28. user (2026-06-22T09:34:12.787Z)

```
=== DRY_RUN location ===
docs/DRY_RUN_CUJ.md
=== mahadevpura_smoke staged? (can't verify GCS, note) ===
(GCS bucket path ? unverifiable locally)
=== screencast paths referenced ===
3
```

29. assistant (2026-06-22T09:34:21.778Z)

I have everything I need. Let me compile the findings.

Summary of verification:

- **SPA back-half + front-half (B-P2)****: All exist. `IngestPanel.tsx`, `useIngestJob.ts`, api client `processVideo`/`getJob`/`/api/query`/`/api/compile`/`/api/highlights` ? verified in `web/src/`. B-P2 landed in #56 (2026-06-21), matching \$9.
- **Engine PRs #280/#281/#289****: All exist in the engine repo with matching descriptions.
- **Engine file:line claims****: Substance verified (`resolve\_input\_ref`, non-local-skips-`os.path.exists` gate at `main.py:590`, `gs://`?`gcs` at `ref.py:60`). Exact line numbers 159/274/283 have drifted (engine shipped more PRs), but the routing logic is intact.
- **Engine test names****: All 4 cited tests exist in `tests/test\_api\_endpoints.py`.
- **Multilingual "en/kn/hi"*****: STALE ? SPA now supports 6 languages (en/kn/hi/es/da/id) per `web/src/i18n/index.ts` and #87.
- **Doc status `DRAFT`****: STALE ? work is DELIVERED per `docs/README.md`; DoD boxes 1-3 done, box 4 parked.

30. user (2026-06-22T09:34:52.817Z)

Structured output provided successfully

31. assistant (2026-06-22T09:35:02.251Z)

Audit complete. I verified `docs/DRY\_RUN\_CUJ.md` against the live khelsutra repo (master at #94) and the sibling engine repo.

Verdict: minor-staleness. The doc's substantive engineering claims hold up well ? the SPA back-half and the B-P2 front-half ingest screen exist (`web/src/components/IngestPanel.tsx`, `web/src/hooks/useIngestJob.ts`, shipped in #56), the engine bucket-ingest routing is verified in code (`backend/main.py:581/590`, `backend/storage/ref.py:60`), and all three cited engine PRs (#280/#281/#289) plus all four cited engine tests exist.

Two stale items worth fixing:

- The header still reads `Status: DRAFT ? owner-gated · Goal for this session`, but `docs/README.md:4` correctly calls the work DELIVERED with DoD boxes 1-3 done (medium).
- The doc repeatedly frames the multilingual workflow as "en/kn/hi" (3 langs), but the SPA now supports 6 (`web/src/i18n/index.ts:25` ? en/kn/hi/es/da/id, es/da/id added in #87) (low).

Minor drift: the engine `file:line` numbers (159/274/283) have shifted as the engine shipped more PRs, though the logic they point at is intact. Run-result artifacts (rally counts, GCS screencast paths) are not locally verifiable but have no counter-evidence.

This is a completed tracker and an archive candidate once the owner-parked droplet box (\$6 box 4) closes.